RAG Learning

1.9 Differentiation

Mathematics B (MEI) — OCR B — A-Level (H640) — Pure Mathematics — spec refs c1–c19 (Calculus)

What this topic covers. Differentiation from first principles, the standard derivatives, the product, quotient and chain rules, implicit and parametric differentiation, and applications — tangents and normals, stationary points, rates of change and optimisation.

1. The derivative

$\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is the **rate of change** of y with respect to x — geometrically, the gradient of the tangent to the curve.

From first principles

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Worked example. Differentiate $y = x^2$ from first principles.

$$\frac{(x+h)^2 - x^2}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h$$

Letting $h \to 0$ gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2x$.

Cancel the h **before** taking the limit. Substituting $h = 0$ into the fraction first gives $\frac{0}{0}$, which is meaningless — the cancellation is the whole point of the method.

2. Standard derivatives

y	$\frac{\mathrm{d}y}{\mathrm{d}x}$
x^n	nx^{n-1}
constant	0
e^x	e^x
e^{kx}	$k\mathrm{e}^{kx}$
$\ln x$	$\frac{1}{x}$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

The trigonometric derivatives hold only when x is in **radians**. This follows from $\lim_{\theta \to 0} \frac{\sin \theta}{\theta} = 1$, which itself requires radians.

Before differentiating, rewrite roots and fractions as powers: $\sqrt{x} = x^{1/2}$ differentiates to $\frac{1}{2}x^{-1/2}$; $\frac{1}{x} = x^{-1}$ differentiates to $-x^{-2}$.

3. The three rules

Chain rule

$$\frac{\mathrm{d}}{\mathrm{d}x} f\big(g(x)\big) = f'\big(g(x)\big) \cdot g'(x)$$

Differentiate $\sin 3x$: outer derivative $\cos 3x$, inner derivative 3, giving $3 \cos 3x$.
Differentiate $(2x+1)^5$: $5(2x+1)^4 \times 2 = 10(2x+1)^4$.

Differentiate e^{x^2}: $e^{x^2} \times 2x = 2xe^{x^2}$.
Differentiate $\ln(2x+1)$: $\dfrac{1}{2x+1} \times 2 = \dfrac{2}{2x+1}$.

Product rule

$$\frac{d}{dx}(uv) = u'v + uv'$$

Worked example. Differentiate x^2e^x.
With $u = x^2$, $v = e^x$: $u' = 2x$, $v' = e^x$, so

$$\frac{dy}{dx} = 2xe^x + x^2e^x = xe^x(2+x)$$

Worked example. Differentiate $x \ln x$.

$$1 \times \ln x + x \times \tfrac{1}{x} = \ln x + 1$$

Quotient rule

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

Worked example. Differentiate $\dfrac{x+1}{x-1}$.

$$\frac{1(x-1) - (x+1)(1)}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

The quotient rule has a **minus** and the order matters: $u'v - uv'$, not $uv' - u'v$. Getting them the wrong way round flips the sign of every answer.

4. Implicit differentiation

When y cannot be made the subject, differentiate both sides with respect to x, applying the chain rule to every y term so that each contributes a factor $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

Worked example. For $x^2 + y^2 = 25$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

$$2x + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \implies \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{x}{y}$$

Worked example. For $x^3 + y^3 = 6xy$, find $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

The right-hand side needs the **product rule**:

$$3x^2 + 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} = 6y + 6x\frac{\mathrm{d}y}{\mathrm{d}x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}\left(3y^2 - 6x\right) = 6y - 3x^2 \implies \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2y - x^2}{y^2 - 2x}$$

5. Parametric differentiation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}$$

Worked example. For $x = t^2$, $y = t^3$: $\dfrac{\mathrm{d}x}{\mathrm{d}t} = 2t$, $\dfrac{\mathrm{d}y}{\mathrm{d}t} = 3t^2$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{3t^2}{2t} = \dfrac{3t}{2}$.

For the **second** derivative you cannot simply differentiate again with respect to t. Use

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) \div \frac{\mathrm{d}x}{\mathrm{d}t}$$

6. Tangents, normals and stationary points

At a point where the gradient is m, the **tangent** has gradient m and the **normal** has gradient $-\frac{1}{m}$.

Worked example. Find the tangent to $y = x^2$ at $(2, 4)$.
$\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2x = 4$, so $y - 4 = 4(x - 2)$, i.e. $y = 4x - 4$.

Classifying stationary points

Stationary points occur where $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$. Then:

Second derivative	Nature
$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} > 0$	Minimum
$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} < 0$	Maximum
$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$	Inconclusive — test the sign of the gradient either side

Worked example. Find and classify the stationary points of $y = x^3 - 3x$.

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 3x^2 - 3 = 0 \implies x = \pm 1$$

Points $(1, -2)$ and $(-1, 2)$. Since $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 6x$, at $x = 1$ it is $6 > 0$ (minimum) and at $x = -1$ it is $-6 < 0$ (maximum).

A zero second derivative does **not** prove a point of inflection. For $y = x^4$ at the origin, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ yet the point is a minimum. A genuine inflection needs the second derivative to *change sign*.

7. Connected rates of change and optimisation

Rates link through the chain rule, for example $\dfrac{\mathrm{d}V}{\mathrm{d}t} = \dfrac{\mathrm{d}V}{\mathrm{d}r} \times \dfrac{\mathrm{d}r}{\mathrm{d}t}$.

Worked example. A sphere has $V = \frac{4}{3}\pi r^3$ and $\dfrac{\mathrm{d}r}{\mathrm{d}t} = 2$. Find $\dfrac{\mathrm{d}V}{\mathrm{d}t}$.

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 4\pi r^2 \implies \frac{\mathrm{d}V}{\mathrm{d}t} = 4\pi r^2 \times 2 = 8\pi r^2$$

For an optimisation problem: write the quantity to be optimised in terms of **one** variable using any constraint, differentiate, set to zero, and justify the nature of the stationary point.

Optimisation answers routinely lose the last mark for not **justifying** that the stationary point is the maximum or minimum required, and for not answering the question actually asked (often the value of the quantity, not the value of x).

8. Common pitfalls

- Substituting $h = 0$ before cancelling in a first-principles derivation.
- Reversing the order in the quotient rule.
- Forgetting the inner derivative in the chain rule.
- Omitting $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ when differentiating a y term implicitly.
- Missing the product rule on a term like $6xy$ in an implicit equation.
- Inverting the parametric fraction.
- Differentiating trigonometric functions with the calculator or the working in degrees.
- Concluding "point of inflection" from a zero second derivative alone.